

CMIP6数据介绍与应用

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主要内容

- CMIP6介绍
- 子比较计划介绍
- 降尺度的方法
- 预备知识1-环境的部署
- 预备知识2-处理
- 统计降尺度-偏差订正
- 动力降尺度-数值预报模式WRF模拟
- 未来气候数据应用- (bgc和swat)

CMIP6介绍

- 简介:

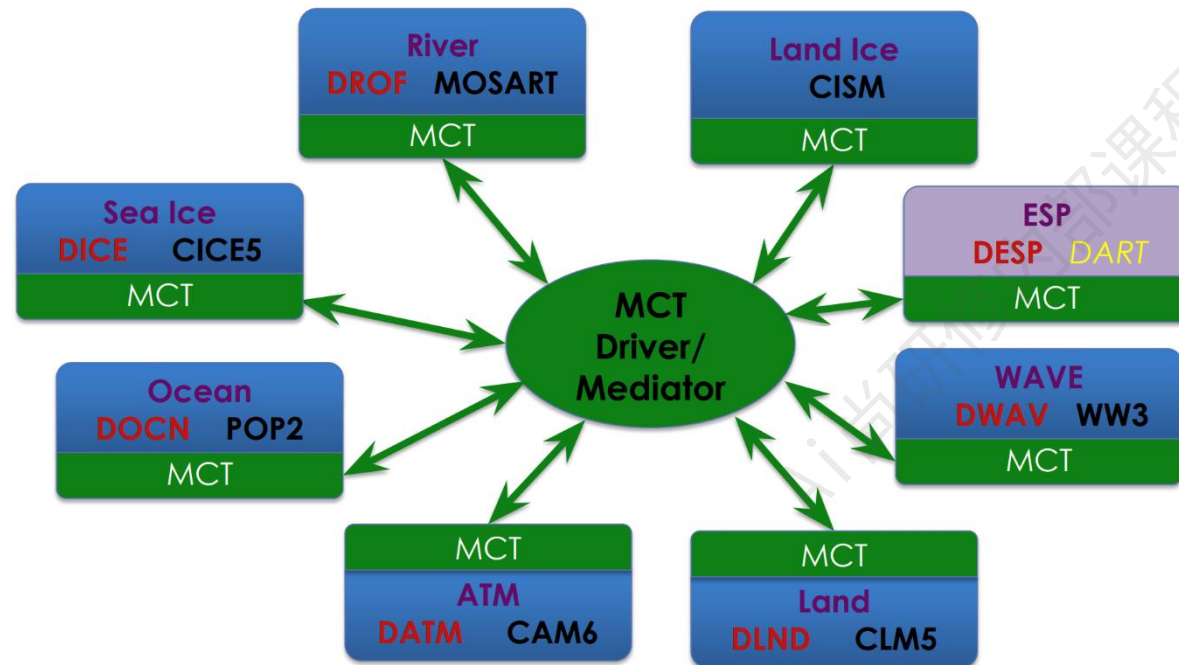
国际耦合模式比较计划 (CMIP), 其基础和雏形为大气模式比较计划 (AMIP) (1989—1994年), 由世界气候研究计划 (WCRP) 耦合模拟工作组 (WGCM) 于 1995 年发起和组织, 其最初目的是对当时数量有限的全球耦合气候模式的性能进行比较。

- 目的:

推动模式发展和增进对地球气候系统的科学理解

常见的GCM-美国CESM模型

Current CESM2 Coupling – data components permit flexible activation/deactivation of feedbacks



MCT – Model Coupling Toolkit



Community
Earth
System
Model



Daily scale

```
cat@node0:~/bigdisk/cmip$ ls
```

```
ACCESS-ESM1-5  CanESM5      CMCC-ESM2    CNRM-ESM2-1  INM-CM5-0    MIROC6       NorESM2-LM  
BCC-CSM2-MR    CESM2-WACCM  CNRM-CM6-1  INM-CM4-8    IPSL-CM6A-LR MRI-ESM2-0    NorESM2-MM
```

```
cat@node0:~/bigdisk/cmip$ █
```

```
-----  
cat@node0:~/bigdisk/cmip$ du -lh -d 0  
1.8T  .
```

```
BCC-CSM2-MR    CESM2-WACCM  CNRM-CM6-1  INM-  
cat@node0:~/bigdisk/cmip$ ls MIROC6/  
historical  ssp126  ssp245  ssp370  ssp585
```

```
-----  
cat@node0:~/bigdisk/cmip$ ls MIROC6/historical/  
hurs  huss  pr  psl  rsds  tas  tasmax  tasmin  uas  vas  
cat@node0:~/bigdisk/cmip$ █
```

Monthly scale

ACCESS-CM2	CAS-ESM2-0	E3SM-1-1-ECA	GISS-E2-1-G	MIROC6	NorESM2-MM
ACCESS-ESM1-5	CESM2	EC-Earth3	GISS-E2-1-H	MPI-ESM-1-2-HAM	SAM0-UNICON
AWI-CM-1-1-MR	CESM2-FV2	EC-Earth3-Veg	IITM-ESM	MPI-ESM1-2-HR	TaiESM1
AWI-ESM-1-1-LR	CESM2-WACCM	EC-Earth3-Veg-LR	INM-CM4-8	MPI-ESM1-2-LR	
BCC-CSM2-MR	CESM2-WACCM-FV2	FGOALS-f3-L	INM-CM5-0	MRI-ESM2-0	
BCC-ESM1	CIESM	FGOALS-g3	IPSL-CM6A-LR	NESM3	
CAMS-CSM1-0	CMCC-CM2-HR4	FIO-ESM-2-0	KACE-1-0-G	NorCPM1	
CanESM5	CMCC-CM2-SR5	GFDL-ESM4	KIOST-ESM	NorESM2-LM	

子比较计划介绍



子比较计划介绍

I. Overview CMIP6-Endorsed MIPs

1. AerChemMIP- Aerosols and Chemistry Model Intercomparison Project
2. C⁴MIP - Coupled Climate Carbon Cycle Model Intercomparison Project
3. CDRMIP - The Carbon Dioxide Removal Model Intercomparison Project
4. CFMIP - Cloud Feedback Model Intercomparison Project
5. DAMIP - Detection and Attribution Model Intercomparison Project
6. DCP - Decadal Climate Prediction Project
7. FAFMIP - Flux-Anomaly-Forced Model Intercomparison Project
8. GeoMIP - Geoengineering Model Intercomparison Project
9. GMMIP - Global Monsoons Model Intercomparison Project
10. HighResMIP - High-Resolution Model Intercomparison Project
11. ISMIP6 - Ice Sheet Model Intercomparison Project for CMIP6
12. LS3MIP - Land Surface, Snow and Soil Moisture
13. LUMIP - Land-Use Model Intercomparison Project
14. OMIP - Ocean Model Intercomparison Project
15. PAMIP - Polar Amplification Model Intercomparison Project
16. PMIP - Palaeoclimate Modelling Intercomparison Project
17. RFMIP - Radiative Forcing Model Intercomparison Project
18. ScenarioMIP - Scenario Model Intercomparison Project
19. VolMIP - Volcanic Forcings Model Intercomparison Project
20. CORDEX - Coordinated Regional Climate Downscaling Experiment
21. DynVarMIP - Dynamics and Variability Model Intercomparison Project
22. SIMIP - Sea Ice Model Intercomparison Project
23. VIACS AB - Vulnerability, Impacts, Adaptation and Climate Services Advisory Board

表 1 CMIP6 委员会批准的 MIPs

Table 1 The CMIP6-endorsed MIPs

序号	模式比较计划	参与的模 式组数
1	气溶胶和化学模式比较计划 (AerChemMIP)	13
2	耦合气候碳循环比较计划 (C4MIP)	19
3	二氧化碳移除模式比较计划 (CDRMIP)	12
4	云反馈模式比较计划 (CFMIP)	19
5	检测归因模式比较计划 (DAMIP)	14
6	年代际气候预测计划 (DCPP)	19
7	通量距平强迫模式比较计划 (FAFMIP)	10
8	地球工程模式比较计划 (GeoMIP)	10
9	全球季风模式比较计划 (GMMIP)	21
10	高分辨率模式比较计划 (HighResMIP)	15
11	冰盖模式比较计划 (ISMIP6)	11
12	陆面、雪和土壤湿度模式比较计划 (LS3MIP)	13
13	土地利用模式比较计划 (LUMIP)	13
14	海洋模式比较计划 (OMIP)	21
15	极地放大模式比较计划 (PAMIP)	10
16	古气候模式比较计划 (PMIP)	14
17	辐射强迫模式比较计划 (RFMIP)	11
18	情景模式比较计划 (ScenarioMIP)	23
19	火山强迫的气候响应模拟比较计划 (VolMIP)	11
20	协同区域气候降尺度试验 (CORDEX)	13
21	平流层和对流层系统的动力学和变率 (DynVarMIP)	13
22	海冰模式比较计划 (SIMIP)	17
23	脆弱性、影响和气候服务咨询 (VIACS AB)	4

降尺度的方法

Ai尚研修内部课程、禁止转载

降尺度的方法

- 统计降尺度
 - 偏差订正
 - 多元回归
- 动力降尺度
 - RegCM
 - WRF

Ai尚研修内部课程、禁止转载

格点化观测-1

- PRSIM
- DAYMET
- MT-CLIM

https://prism.oregonstate.edu



Northwest Alliance for Computational Science and Engineering

Home Normals Comparisons This Month Prior 6 Months Recent Years Historical Past Projects Explorer FAQ

PRISM Climate Data

The PRISM Climate Group gathers climate observations from a wide range of monitoring networks, applies sophisticated quality control measures, and develops spatial climate datasets to reveal **short- and long-term climate patterns**. The resulting datasets incorporate a variety of modeling techniques and are available at multiple spatial/temporal resolutions, covering the period from 1895 to the present. Whenever possible, we offer these datasets to the public, either free of charge or for a fee (depending on dataset size/complexity and funding available for the activity).

- Methods used by the [PRISM model](#)
- Descriptions of the [PRISM datasets](#)
- [How we developed the PRISM model](#)
- [Publications about PRISM](#)
- Calendar of PRISM [dataset updates](#)
- [What's new](#) at PRISM
- [PRISM Updates mailing list](#)

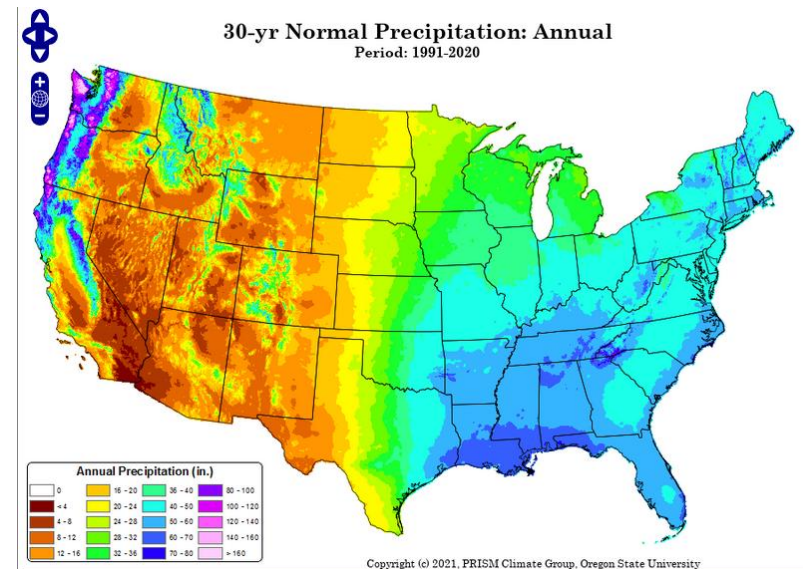
[Click here](#) to learn how to participate in citizen science and help improve PRISM data!

High-Resolution Data Available

The native grid resolution of the PRISM datasets is 800m, but they have been filtered to 4km resolution for easier downloading and manipulation on this website. The 800m versions of all PRISM datasets, which contain 25x more information, are available to users for a fee. Details on availability, pricing, and ordering are found [here](#).

30-Year Normals At the end of each decade, average values for temperature and precipitation are computed over the preceding 30 years. The current set of 30-year normals covers the period 1991-2020.

Comparisons Maps showing how observed values have been deviating from long-term conditions (also known as anomalies) - includes the new Drought Indicator tool.



格点化观测-2

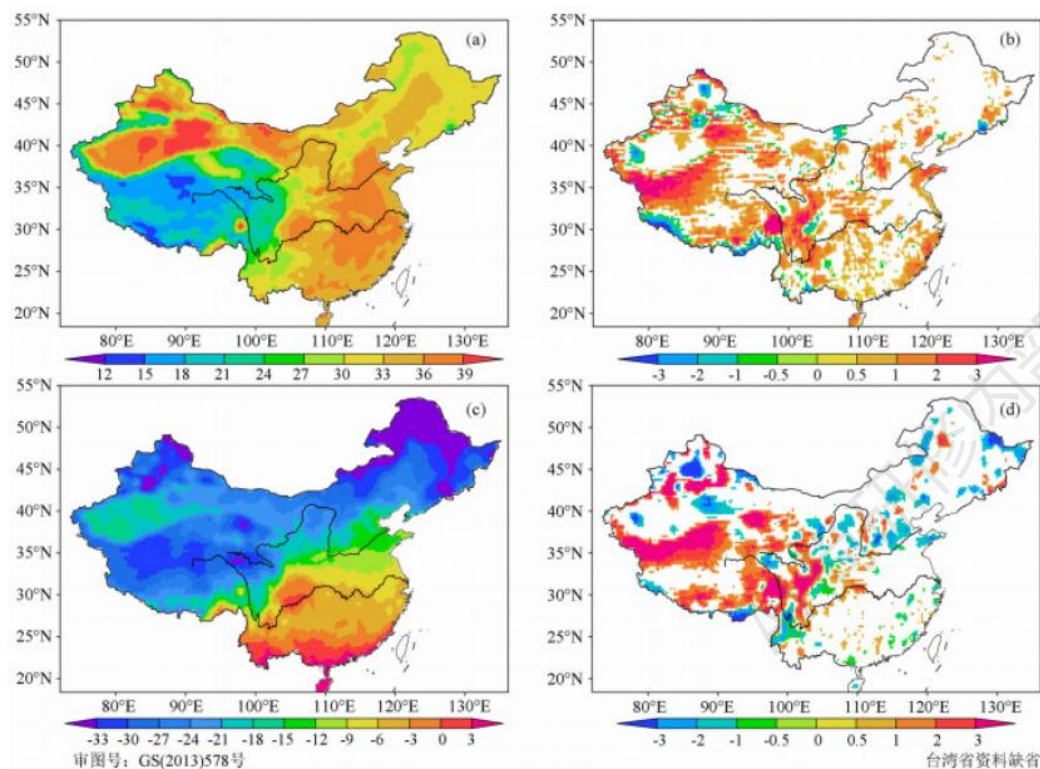
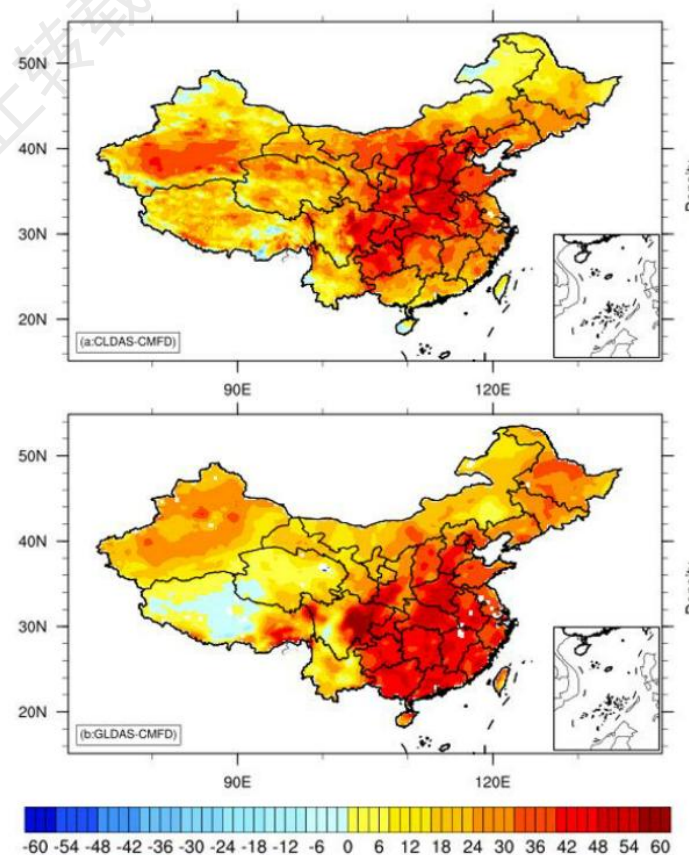


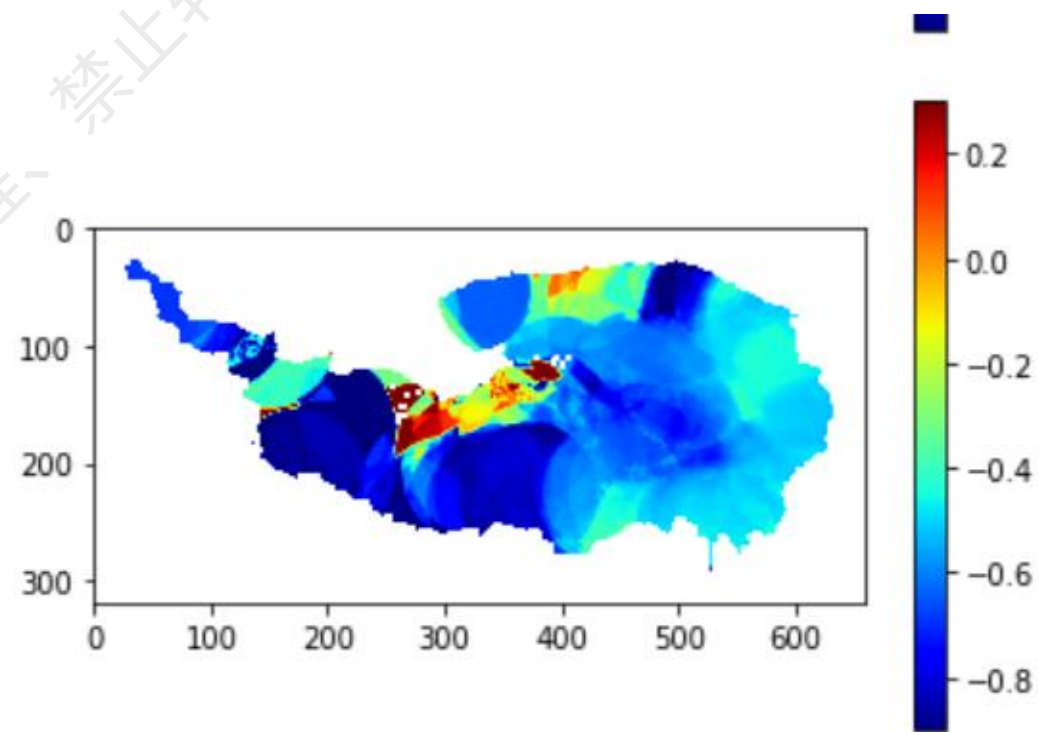
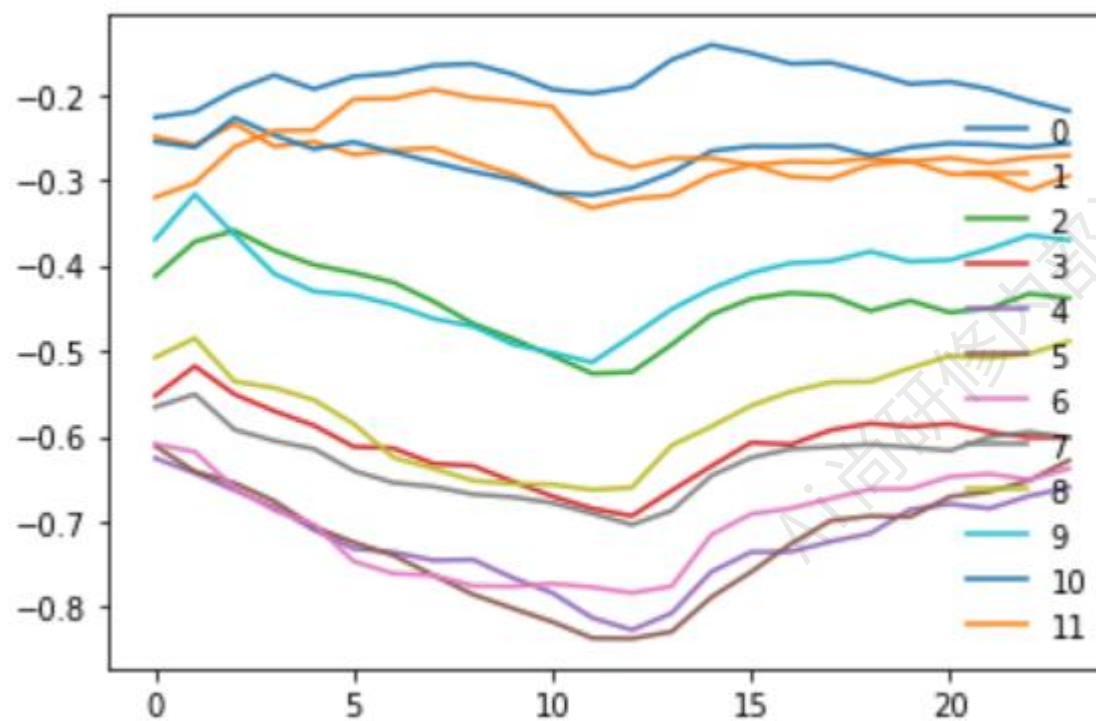
图3 研究区CN05.1的气温极端事件(左侧)及与CN05.1的差(右侧): (a, b) TX3D; (c, d) TN3D (单位: $^{\circ}\text{C}$)

Fig. 3 Extreme temperature indices from CN05.1 (left column) and the difference between it and CN05.1 in the study area: (a, b) TX3D; (c, d) TN3D (units: $^{\circ}\text{C}$)

吴佳,2013



Fan Yang,2017



统计降尺度

- 假设前提：
 - 建立的关系是**不随时间变化、周期性的重复**
 - 误差是系统性
- 思路：
 - 建立GCM与观测之间**映射关系**
- 常见方法：
 - 多元回归
 - 偏差订正
 - 随机天气发生器